# Recovering the primes in a dyadic interval from exact moments: two constructive routes and an obstruction theorem

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**Abstract**

The companion papers supply two exact "moment channels" for the primes in a dyadic window $W = (m, 2m]$: integer power sums $\sum_{p \in W} p^k$, produced from divisor-lattice data with no prime input at any stage [1], and weighted logarithmic moments $\sum_{p \in W} Q(p) \log p$ for polynomial $Q$, produced from the primes below the window [2]. This note closes the circle: the channels determine the window primes *individually*. Two constructive decoders are given. From the $r+1$ integer moments of weights $0, \ldots, r$ ($r$ the number of window primes), Newton's identities yield the elementary symmetric functions — integers — and the primes are read off as the integer roots of the resulting monic polynomial; the entire pipeline contains no prime input and no primality test. Alternatively, a *single* real moment, Chebyshev's 1852 functional $\theta(2m) - \theta(m)$, known to $\lceil \log_2 N \rceil + 2$ bits where $N = \prod_{p \in W} p \leq 4^m$, determines $N$ by rounding, and the window primes are exactly the divisors of $N$ in $W$, found by trial division; a Prony decoder using $2r$ moments at intermediate precision interpolates between the two. Against these, an obstruction: any scheme whose decoder receives at most $D$ bits determined by the configuration, and which correctly decodes *every* $r$-subset of the window's integers, has $D \geq \log_2 \binom{m}{r} - 1 \asymp m \log \log m / \log m$. Hence no uniform scheme of bounded description decodes the dyadic windows for all $m$, while the companion's certified determination of the single aggregate $S(m)$ from $O(\log m)$ moments sits near the same floor read at entropy $O(\log m)$: one pigeonhole, two entropy scales. The model restriction — decoders blind to number theory except through the delivered bits — is discussed and shown to be necessary; Mills- and Willans-type formulas live outside it, as does the sieve. All constructions are machine-verified: at $m = 500$ both routes recover the 73 primes of $(500, 1000]$ exactly, the second from a single 695-bit real number.

**Keywords:** prime recovery; moment problems; Newton's identities; elementary symmetric functions; Chebyshev's functional; Prony's method; information-theoretic lower bounds.

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## 1 Introduction

The first companion paper [1] develops Linnik's identity [3], in the computational tradition opened by McKenzie [4], into a formula system $S_j$ returning the exact integers $\sum_{p \leq y} p^j$ for

*Version 1.0, August 2026. Companion to *Prime counts and prime sums from multiplication tables: a self-similar formula system* [1] and *Certified exact sums of primes over dyadic intervals via a ladder of generalized factorials* [2] (same author). Developed in collaboration with Claude (Anthropic); all numerical claims were machine-verified as described in Section 6. Responsibility for the content rests with the author. A full disclosure of AI use appears at the end of the paper.

every weight $j \geq 0$ from divisor-lattice data alone: no prime, primality test, or sieve appears at any depth. The second [2] constructs generalized factorials $\prod_{i \leq 2m} i^{Q(i)}$ with polynomial exponents whose valuations at window primes are $Q(p)$, so that each polynomial weight yields one exact functional $\sum_{m<p\leq 2m} Q(p) \log p$, computable from the primes $\leq m$; finitely many such moments, closed by a certified sandwich and the integrality of the target, determine the aggregate $S(m) = \sum_{m<p\leq 2m} p$ exactly.

Both constructions output *aggregates*. It is natural to ask how much of the window the moment channels actually know. The answer, made precise here, is: everything. Finitely many moments determine the window primes individually, by elementary and fully constructive decoders (Theorems 1 and 2, Remark 1), and the cost of this determination — counted in delivered bits — can be compared against an unconditional pigeonhole floor (Theorem 3). The comparison quantifies, in a restricted but honestly delimited model, the folklore assertion that no short formula produces the primes: the floor grows like $m \log\log m / \log m$ for the full window, while a single aggregate of entropy $O(\log m)$ admits the companion's $O(\log m)$-moment certified determination. The impossibility of a bounded-length exact scheme for the primes themselves and the possibility of a log-length certified procedure for one sum are the same counting argument read at two entropy scales.

We state claims and non-claims plainly. Claimed: (i) the exact recovery theorems and their machine verification; (ii) the bit-budget lower bound, with an explicit discussion of the model restriction that makes it true; (iii) the accounting that unifies the three regimes (one exact real moment; many finite-precision moments; too few bits). Not claimed: computational advantage — both decoders cost far more than sieving the window, by design, since the object of study is the information channel, not the running time; and no unrestricted impossibility — the primes are computable from $m$, so any lower bound must, and ours does, constrain the decoder's access rather than the universe.

## 2 The two moment channels

Fix $m \geq 8$, write $W = (m, 2m]$, let $P(W)$ be the set of primes in $W$ and $r = |P(W)|$ (so $r \geq 1$ by Bertrand's postulate). The channels of [1, 2] deliver:

$$\text{(integer channel)} \qquad s_k \;=\; \sum_{p\in P(W)} p^k \;=\; S_k(2m) - S_k(m) \;\in\; \mathbb{Z}, \qquad k \geq 0, \tag{1}$$

$$\text{(weighted channel)} \qquad M_Q \;=\; \sum_{p\in P(W)} Q(p)\,\log p, \qquad Q \in \mathbb{Z}[t], \tag{2}$$

the first exactly, in integer arithmetic, with no prime input at any stage [1]; the second to any prescribed precision, from the primes $\leq m$ together with logarithms of the integers $\leq 2m$ [2, Prop. 1]. The case $Q = 1$ of (2) is Chebyshev's identity [5] in the form $\theta(2m) - \theta(m) = \log\,(2m)! - \sum_{p\leq m} v_p\big((2m)!\big) \log p$, used below. Only the two displayed facts (1) and (2), with the stated computability provenance, are imported from the companions; every proof in this paper is self-contained given those displays.

## 3 Route I: integer moments, Newton's identities, integer roots

**Theorem 1** (Prime-free individual recovery). *Let $r = s_0 = S_0(2m) - S_0(m)$ and let $s_1, \ldots, s_r$ be as in* (1). *Define $e_0 = 1$ and, recursively,*

$$e_k \;=\; \frac{1}{k}\sum_{i=1}^{k}(-1)^{i-1}e_{k-i}\,s_i, \qquad 1 \le k \le r.$$

*Then every $e_k$ is an integer,*

$$f(x) \;=\; \sum_{k=0}^{r}(-1)^k e_k\,x^{r-k} \;=\; \prod_{p\in P(W)}(x-p),$$

*and $P(W) = \{\, c \in W \cap \mathbb{Z} : f(c) = 0 \,\}$. The entire pipeline — moments, recursion, and root reading by evaluation at the $m$ integers of $W$ — consumes no prime and performs no primality test; the primes emerge as the integer roots.*

*Proof.* The $s_k$ are the power sums of the finite multiset $P(W)$ of distinct integers. Newton's identities (see, e.g., [6, Ch. I, §2]) express the elementary symmetric functions $e_k$ of that multiset by exactly the stated recursion; the $e_k$ are symmetric polynomials with integer coefficients in integers, hence integers, and the recursion's divisions by $k$ are exact. The monic polynomial with those symmetric functions is $\prod_p(x-p)$. A monic integer polynomial of degree $r$ has at most $r$ integer roots; the $r$ window primes are roots; so the zero set of $f$ in $W$ is exactly $P(W)$. □

Recovery of a finite multiset from its power sums is classical and has a modern general theory [12]; in exact arithmetic, the same decoder — power sums, a Hankel/Newton step, integer roots — is the sparse-interpolation algorithm of Ben-Or and Tiwari [13]. What is specific here is the provenance and the support: the power sums arrive from a prime-free channel, and the roots are the window primes.

The moment count is the smallest for which this decoder is defined: $r$ unknowns, $r$ power sums, plus $s_0$ to know $r$. The price is paid in height rather than count: $\log_2 e_k$ grows like $k\log_2 m$, so the total delivered data is $\Theta(r^2\log m)$ bits (Section 7), and the encoder must run the companion system at weights up to $r$.

## 4 Route II: one real moment, rounding, trial division

**Theorem 2** (Single-moment recovery). *Let $N = \prod_{p\in P(W)} p$, so $N \mid \binom{2m}{m}$ and $N \le 4^m$. Put $B = \lceil \log_2 N\rceil + 2$ (a priori, $B = 2m+2$ suffices). Let $\widetilde{M}$ be any real number with*

$$\bigl|\widetilde{M} - \bigl(\theta(2m) - \theta(m)\bigr)\bigr| \;\le\; 2^{-B},$$

*for instance the output of the extraction identity of [2, Prop. 1] at rung 0, evaluated in interval arithmetic at $B+O(1)$ bits from the primes $\le m$ and the logarithms of the integers $\le 2m$. Then $N$ is the unique integer within distance $\frac{1}{2}$ of $\exp(\widetilde{M})$, and*

$$P(W) \;=\; \{\, c \in W \cap \mathbb{Z} : c \mid N \,\},$$

*recovered by trial division of $N$ by the $m$ integers of the window. No integer in $W$ is tested for primality.*

*Proof.* $\exp(\theta(2m) - \theta(m)) = N$ exactly, and for $|\delta| \le 2^{-B}$, $|Ne^{\delta} - N| \le N\,2^{-B+1} < 2^{\lceil \log_2 N\rceil+1-B} = \frac{1}{2}$, so rounding recovers $N$; the divisibility $N \mid \binom{2m}{m}$ and the bound $N \le 4^m$ are the central binomial facts underlying [5, 10]. For the divisor characterization: every $p \in P(W)$ divides $N$ and lies in $W$; conversely a composite $c \in W$ has a prime factor $q \le \sqrt{2m} \le m$ (valid for $m \ge 2$), and $q$ does not divide $N$ since every prime factor of $N$ exceeds $m$; hence $c \nmid N$. □

Theorem 2 is the "internal Willans" of the moment model: a single real number carries the entire window. Two comments calibrate it. First, the information is priced in precision — the one moment must be delivered to $\approx \log_2 N \approx (\theta(2m) - \theta(m))/\log 2$ bits, and Section 7 shows this is within a factor $\sim \log m/\log\log m$ of the unconditional floor, making rung 0 at high precision the most bit-efficient constructive decoder we know. Second, information and extraction separate: the *exact* real $\theta(2m) - \theta(m)$ determines $P(W)$ outright, but generic extraction from $N$ is integer factorization of an $r \log m$-bit number; the window structure rescues us, since trial division over $W$ is *complete* — composites cannot divide $N$ — so no factoring and no primality testing occur.

The kernel of Route II — the primorial determines the primes — has classical avatars. Gandhi's formula recovers the next prime from the divisors of $p_n\#$ by a Möbius sum [14, 15], and Fridman, Garbulsky, Glecer, Grime, and Tron Florentin exhibited a single real constant, in effect a primorial-denominated series, whose floor recurrence emits every prime [16]. Both are decoders whose data store the answers, and both therefore live outside the model of Section 5, exactly as Mills's constant does; Theorem 2 differs in taking the real number from a computable moment channel and pricing the window's information in delivered bits.

*Remark* 1 (Prony interpolation). The weighted channel (2) with $2r$ moments recovers $P(W)$ at intermediate precision. With $\sigma_k = \sum_p (\log p)\, p^k$ for $0 \le k \le 2r-1$, the Hankel matrix $(\sigma_{i+j})_{0\le i,j\le r-1} = V^{\mathsf{T}} D V$ is positive definite ($V$ the Vandermonde matrix at the distinct primes, $D = \operatorname{diag}(\log p) \succ 0$), and the Prony system $\sum_{j=0}^{r-1} c_j \sigma_{i+j} = -\sigma_{r+i}$ returns the coefficients of $\prod_p (x-p) \in \mathbb{Z}[x]$, which rounding makes exact at sufficient precision; the roots are then read off as in Theorem 1. Explicit precision bounds follow from Vandermonde-inverse estimates using the gap $|p_i - p_j| \ge 2|i-j|$ between distinct odd primes, but are tedious; sharp forms exist in the super-resolution literature, where the required precision is governed by node separation [26, 27], and the primes' gap $\ge 2$ places the window in the well-separated regime. We record instead the machine datum that at $m = 50$ a working precision of about 160 bits suffices (Section 6). Prony's memoir is [7].

## 5 The obstruction theorem

The primes up to $2m$ are computable from the single input $m$; in the unrestricted world their description length is $O(\log m)$ and no lower bound is possible. The meaningful question, and the rigorous kernel of the no-closed-form folklore (classically articulated by Wilf and Dudley [19, 20]), concerns schemes whose decoders learn about the window *only through delivered data.* We formalize the channel and count.

**Definition 1** (Uniform scheme)**.** *Fix $m$ and $1 \le r \le m$. A $D$-bit* uniform scheme *for $(m, r)$ is a pair $(\Phi, \Psi)$ where $\Phi$ maps each $r$-element subset $S \subseteq W \cap \mathbb{Z}$ to a bit string of length at most $D$, and $\Psi(\Phi(S)) = S$ for every such $S$. No constraint is placed on the functions themselves: $\Phi$ may be any quantization of any moments, and $\Psi$ any decoder — but $\Psi$ sees only the bits.*

**Theorem 3** (Bit floor)**.** *Every $D$-bit uniform scheme for $(m, r)$ satisfies $D \geq \log_2 \binom{m}{r} - 1$. In particular, for $r = \pi(2m) - \pi(m) \sim m/\log m$,*

$$D \;\geq\; (1 + o(1))\, \frac{m \log_2(e \log m)}{\log m} \;\longrightarrow\; \infty,$$

*so no family of uniform schemes of bounded description length decodes the dyadic windows for all $m$.*

*Proof.* $\Psi \circ \Phi = \mathrm{id}$ forces $\Phi$ to be injective on the $\binom{m}{r}$ configurations; there are $2^{D+1} - 1$ bit strings of length at most $D$; so $2^{D+1} > \binom{m}{r}$. The asymptotic is $\log_2 \binom{m}{r} = r \log_2(em/r) + O(\log m)$ with $r \sim m/\log m$. $\square$

The counting step is the standard information-theoretic floor for representing $r$-subsets of an $m$-universe, as in succinct data structures [24] and combinatorial search [25]; the content here is its application to the moment channels and the accounting of Section 7. We note also that "entropy of the primes" has an established different use — entropy *proofs* of Chebyshev-type estimates [23] — distinct from the subset-counting entropy priced here; and that the nearest prior impossibility statement we know, an algorithmic-information argument against "formulas for primes" [22], is heuristic and asymptotic in character, where Theorem 3 is a finite, unconditional count in an explicitly delimited channel model.

Three glosses carry the theorem's meaning. *(a) The ensemble is the model.* The actual window primes are one configuration per $m$; correctness over all $r$-subsets is what excludes decoders with number theory built in. Mills's prime-representing function [8] is a decoder whose constant stores the answers — computing the constant requires knowing the primes it emits [21]; Willans's formula [9] passes unbounded information through the evaluation of factorials under a floor, as do its Diophantine and arithmetic-term descendants [17, 18]; a sieve is a decoder that ignores the channel and recomputes from $m$. All of these are outside Definition 1, and their exclusion is precisely what makes the folklore well-posed rather than false. The decoders of Theorems 1 and 2 — Newton's recursion, rounding, trial division — use no property of the primes beyond what the bits assert, are correct on arbitrary configurations of distinct window integers (for Theorem 2, arbitrary sets of primes in $W$; for Theorem 1, arbitrary distinct integers), and hence live inside the model, where their budgets can be compared to the floor. *(b) Precision must be charged.* With exact reals even one moment suffices (Theorem 2), so a count-only lower bound is impossible; the bit budget $D$ is the honest currency, and the count–precision *tradeoff* is the object of study. *(c) Information is not extraction.* The floor prices what the bits can distinguish, not the work of deciding; the time cost of prime determination is a separate and harder question, on which the elementary $\sqrt{x}$-barrier literature [11] is the state of the art. That work is also, in the present vocabulary, a moment channel with a partial decoder: it extracts one exact bit of the window — the parity of the number of its primes — in time $O(N^{1/2-c})$, without enumerating them.

# 6 Machine verification

A reference implementation (`prime_recovery.py` in the code repository [28]) exercises both routes end to end, re-using the companion systems as published. At $m = 500$, so $W = (500, 1000]$: the integer channel returned $r = s_0 = 73$ and $s_1, \ldots, s_{73}$ with no prime input at any stage (the companion system's exactness assertion — every harmonic combination landing on denominator 1 — held at all weights up to 73); Newton's recursion divided exactly at every step; the 73 integer roots of $f$ in $W$ coincide with an independent sieve; and $e_{73} = N$, a 695-bit integer, all in under a second. Independently, rung 0 evaluated by the extraction identity at 1300-bit working precision, from the primes $\leq 500$ and the logarithms of the integers $\leq 1000$, satisfied $|\exp(\widetilde{M}) - N| \approx 4 \times 10^{-178}$; rounding recovered the same $N$ (the cross-route identity $e_{73} = N$ is itself a check that the two channels agree), and trial division over the window returned exactly the 73 primes, with no primality test performed. At $m = 50$, the Prony decoder of Remark 1 with $2r = 20$ weighted moments recovered the ten primes of $(50, 100]$; a precision scan located the working threshold near 160 bits, every rounded coefficient landing within 1/4 of the integer coefficients of $\prod_p (x - p)$. Trust base: big-integer arithmetic, the mpmath library, and Newton's identities; nothing else is assumed.

# 7 The accounting, the gap, and open questions

Theorem 3 supplies the floor; the constructions supply budgets. At the verified scales:

| scheme | window | moments | total delivered bits | $\log_2 \binom{m}{r}$ |
|---|---|---|---|---|
| Route I (Thm. 1) | $(500, 1000]$ | 74 | 27,128 | $\approx 296$ |
| Route II (Thm. 2)$^\dagger$ | $(500, 1000]$ | 1 | 697 | $\approx 296$ |
| Prony (Rem. 1)$^\dagger$ | $(50, 100]$ | 20 | $\approx$ 3,200 | $\approx 33$ |

Route I is uniform over arbitrary distinct-integer configurations and so lies fully inside Definition 1; its $\times 90$ overpay is the redundancy of the symmetric-function encoding, $\Theta(r^2 \log m)$ bits against a floor of $\Theta(r \log\log m)$ — the price of the minimal moment *count*. The daggered schemes are uniform over configurations of window *primes* only: over arbitrary integer configurations a single moment cannot even separate — $\{54, 96\}$ and $\{64, 81\}$ share their product — so the injectivity of rung 0 on prime configurations is itself a unique-factorization fact, and the 697-against-296 comparison bounds the overpay of the product encoding *granted* the side information that the configuration is prime. That overpay has a clean shape: the product spends $\log_2 p \approx \log_2 m$ bits per prime where the floor charges only $\log_2(em/r) \approx \log_2(e \log m)$, a ratio $\log m / \log(e \log m)$, about 2.4 at $m = 500$ and slowly divergent. Rung 0 at high precision is thus the most bit-efficient constructive decoder we know, and it is not optimal.

The unification promised in the introduction is now one sentence. The companion sandwich [2] determines the single aggregate $S(m)$ — entropy $O(\log m)$ — from $K + 1 \asymp \log m$ moments of bounded precision, a polylogarithmic budget sitting near the floor at the small entropy scale; Theorem 3 shows the full window — entropy $\asymp m \log\log m / \log m$ — admits no bounded budget at all. The impossibility of a short exact scheme for the primes themselves and the existence of a log-length certified procedure for one sum are the same pigeonhole read at two entropy scales, which is, we believe, the honest rigorous shape of the no-closed-form folklore: no more, and no less.

Three questions remain open. (1) *The gap.* Does any scheme, uniform over prime configurations, decode the window from $(1 + o(1)) \log_2 \binom{m}{r}$ bits? We conjecture that smooth polynomial-weight moments cannot close the $\log m / \log\log m$ factor, and we know no encoding that does. (2) *Adaptivity.* Extending Theorem 3 to schemes that adaptively query real functionals at bounded precision is routine by the same counting applied to the transcript; a version with *analog* noise models may be less routine. (3) *Time.* The floor prices distinguishability, not work: whether the primes of $W$ can be *found* substantially faster than the elementary square-root barrier is the standing problem of [11], and nothing here touches it. The three regimes — one exact real moment (information complete, extraction hard); many finite-precision moments (constructive, overpaying); too few bits (impossible) — delimit what the moment channels of [1, 2] can and cannot know, and close the circle this series of notes set out to draw.

## Data availability

The complete code underlying this study, including the script that reproduces every recovery experiment reported in Section 6 (`prime_recovery.py`, driven by `run_all.py`), is openly available in the repository [28] under the MIT license. No external datasets were used; all results are generated deterministically by the published code.

## Disclosure of AI use

In accordance with publisher policies on the use of artificial intelligence, the author discloses the following. This paper was developed in an extensive collaboration with Claude (Anthropic; Claude Fable 5, accessed through the claude.ai interface, 2026). The direction of the work — the research programme of exact moment systems for interval primes — together with all decisions about scope, claims, and submission, are the author's. Within that direction, the AI system contributed substantially to the mathematical development and proofs, wrote the verification code, drafted and revised the manuscript text, and assisted with literature search; the reason for its use was to enable a single non-specialist author to develop, check, and document the material to a publishable standard. Every numerical and mathematical claim was subsequently machine-verified by the recovery experiments reported in Section 6, with the verification code openly available in the cited repository [28]. The author has reviewed the entire manuscript and takes full responsibility for its content, including the accuracy of the mathematics and of all references.

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